

Massless Spectra from Cosmic String Cusps

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1 Cosmic Strings

The evolution of an one-dimensional cosmic string in the limit of thin string width can be modelled by the Nambu-Goto action

$$S = -\mu \int d^2\zeta \sqrt{-\gamma}$$

where $\gamma = \det \gamma^{ab}$ and

$$\gamma_{ab} = \partial_a X_\mu \partial_b X^\mu = g_{\mu\nu} \frac{\partial X^\mu}{\partial \zeta^a} \frac{\partial X^\nu}{\partial \zeta^b}$$

The parametrization $\zeta^a = (\zeta^0, \zeta^1) = (\tau, \sigma)$ describes the trajectory of the cosmic strings $X^\mu(\zeta^a)$. We can choose the parameters ζ^a by defining $\dot{X}^\mu(\zeta^a) = \frac{\partial X^\mu}{\partial \tau}$ as a time-like vector and $X'^\mu(\zeta^a) = \frac{\partial X^\mu}{\partial \sigma}$ as a space-like vector.

To derive the equation, we first consider the equivalent form, the Polyakov action

$$S = -\frac{\mu}{2} \int d^2\zeta \sqrt{-h} h^{ab} \gamma_{ab} = -\frac{\mu}{2} \int d^2\zeta \sqrt{-h} h^{ab} g_{\mu\nu} \partial_a X^\mu \partial_b X^\nu$$

where h_{ab} is the metric on the 2-dimensional worldsheet. The relation between h_{ab} and γ_{ab} can be discovered by varying h^{ab}

$$0 = \delta S = -\frac{\mu}{2} \int d^2\zeta \sqrt{-h} \left(-\frac{1}{2} h_{ab} h^{cd} \gamma_{cd} + \gamma_{ab} \right) \delta h^{ab} \quad \Rightarrow \quad \gamma_{ab} = \frac{1}{2} h_{ab} h^{cd} \gamma_{cd}$$

Taking the determinant, we obtain

$$\gamma = \left(\frac{1}{2} h^{ab} \gamma_{ab} \right)^2 h \quad \Rightarrow \quad \sqrt{-h} = \frac{2\sqrt{-\gamma}}{h^{ab} \gamma_{ab}}$$

and that gives

$$S = -\frac{\mu}{2} \int d^2\zeta \sqrt{-h} h^{ab} \gamma_{ab} = -\mu \int d^2\zeta \sqrt{-\gamma}$$

For a Minkowski worldsheet metric $\sqrt{-h} h^{ab} \rightarrow \eta^{ab}$, we can vary the Polyakov action with $X^\mu(\zeta^a)$

$$\begin{aligned} 0 = \delta S &= -\mu \int d^2\zeta \eta^{ab} g_{\mu\nu} \partial_a X^\mu \partial_b (\delta X^\nu) \\ &= \mu \int d^2\zeta \eta^{ab} g_{\mu\nu} \partial_a \partial_b X^\mu \delta X^\nu + (\text{boundary terms}) \end{aligned}$$

Neglecting the boundary terms, consequently, we obtain

$$0 = \eta^{ab} \partial_a \partial_b X^\mu = \ddot{X}^\mu - X''^\mu$$

Additionally, the relation between $h_{ab} = \eta_{ab}$ and γ_{ab} gives the conformal gauge condition.

$$0 = \gamma_{ab} - \frac{1}{2} \eta_{ab} \eta^{cd} \gamma_{cd} = \partial_a X_\mu \partial_b X^\mu - \frac{1}{2} \eta_{ab} (\dot{X}_\mu \dot{X}^\mu - X'_\mu X'^\mu) \Rightarrow \begin{cases} \dot{X}_\mu \dot{X}^\mu + X'_\mu X'^\mu = 0 \\ \dot{X}_\mu X'^\mu = 0 \end{cases}$$

Removing the residual gauge freedom by setting $X^0 = \tau$, we can write a general solution for the string motion as

$$\mathbf{X}(\tau, \sigma) = \frac{1}{2} [\mathbf{X}_+(\sigma_+) + \mathbf{X}_-(\sigma_-)]$$

where $\sigma_\pm = \tau \pm \sigma$. The gauge condition becomes

$$\begin{cases} 1 - \frac{1}{4} |\mathbf{X}_+^{(1)}(\sigma_+) + \mathbf{X}_-^{(1)}(\sigma_-)|^2 - \frac{1}{4} |\mathbf{X}_+^{(1)}(\sigma_+) - \mathbf{X}_-^{(1)}(\sigma_-)|^2 = 0 \\ (\mathbf{X}_+^{(1)}(\sigma_+) + \mathbf{X}_-^{(1)}(\sigma_-)) \cdot (\mathbf{X}_+^{(1)}(\sigma_+) - \mathbf{X}_-^{(1)}(\sigma_-)) = 0 \end{cases} \Rightarrow |\mathbf{X}_\pm^{(1)}(\sigma_\pm)|^2 = 1$$

where $\mathbf{X}_\pm^{(n)} = \frac{d^n \mathbf{X}_\pm(\sigma_\pm)}{d\sigma_\pm^n}$. The evolution of the string can therefore be described by trajectories of $\mathbf{X}_\pm^{(1)}$ on a 2-dimensional unit sphere, known as the Kibble-Turok sphere

2 Cusps

There are certain points on the Nambu-Goto strings that reaches the speed of light. At these points, the parametrization becomes singular. The shape is called a cusp, which occurs whenever

$$|\dot{\mathbf{X}}_c|^2 = 1 \Rightarrow \mathbf{X}'_c = 0 \Rightarrow \mathbf{X}_{c+}^{(1)} = \mathbf{X}_{c-}^{(1)}$$

For a generic cusp, we expand around the cusp point by

$$\mathbf{X}(0, \sigma) = \frac{\sigma^2}{4} (\mathbf{X}_{c+}^{(2)} + \mathbf{X}_{c-}^{(2)}) + \frac{\sigma^3}{12} (\mathbf{X}_{c+}^{(3)} - \mathbf{X}_{c-}^{(3)}) + \mathcal{O}(\sigma^4)$$

where we define the coordinates so the cusp occurs at $\mathbf{X}_c(0, 0) = 0$

Differentiating the condition $|\mathbf{X}_\pm^{(1)}(\sigma_\pm)|^2 = 1$, we obtain

$$\mathbf{X}_\pm^{(1)} \cdot \mathbf{X}_\pm^{(2)} = 0, \quad \mathbf{X}_\pm^{(1)} \cdot \mathbf{X}_\pm^{(3)} = -|\mathbf{X}_\pm^{(2)}|^2$$

3 Einstein Equations for Gravitational Waves

The trace-reversed metric perturbation $\bar{h}^{\mu\nu} = h^{\mu\nu} - \eta^{\mu\nu} h^\lambda{}_\lambda$ is sourced by the stress-energy tensor $T^{\mu\nu}(x)$ through the linearized Einstein equations

$$\square \bar{h}^{\mu\nu} = 4\pi\kappa_g T^{\mu\nu}$$

where $\kappa_g = 4G$ in natural units. To solve the equation, we use

$$\begin{aligned}\bar{h}^{\mu\nu}(t, \mathbf{x}) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t} \tilde{\bar{h}}^{\mu\nu}(\omega, \mathbf{x}) \\ T^{\mu\nu}(t, \mathbf{x}) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega e^{i\omega t} \tilde{T}^{\mu\nu}(\omega, \mathbf{x})\end{aligned}$$

so we solve the Helmholtz equation

$$(\nabla^2 + \omega^2) \tilde{\bar{h}}^{\mu\nu}(\omega, \mathbf{x}) = -4\pi\kappa_g \tilde{T}^{\mu\nu}(\omega, \mathbf{x})$$

Recall that

$$(\nabla^2 + \omega^2) G(\mathbf{x} - \mathbf{x}') = -\delta^{(3)}(\mathbf{x} - \mathbf{x}') \quad \Rightarrow \quad G(\mathbf{x} - \mathbf{x}') = \frac{1}{4\pi} \frac{e^{i\omega|\mathbf{x} - \mathbf{x}'|}}{|\mathbf{x} - \mathbf{x}'|}$$

We can write

$$\tilde{\bar{h}}^{\mu\nu}(\omega, \mathbf{x}) = \kappa_g \int d^3x' \frac{e^{i\omega|\mathbf{x} - \mathbf{x}'|}}{|\mathbf{x} - \mathbf{x}'|} \tilde{T}^{\mu\nu}(\omega, \mathbf{x}')$$

Using the approximation that the source is localized at $\mathbf{x}' = 0$ and is much smaller than the distance $|\mathbf{x}| = r$ to the observer

$$|\mathbf{x} - \mathbf{x}'| = \sqrt{r^2 - 2\mathbf{x} \cdot \mathbf{x}' + x'^2} \approx r - \hat{\mathbf{n}} \cdot \mathbf{x}'$$

where $\hat{\mathbf{n}} = \frac{\mathbf{x}}{r}$, we can write

$$\tilde{\bar{h}}^{\mu\nu}(\omega, \mathbf{x}) \approx \kappa_g \int d^3x' \frac{e^{i\omega(r - \hat{\mathbf{n}} \cdot \mathbf{x}')}}{r} \tilde{T}^{\mu\nu}(\omega, \mathbf{x}') = \kappa_g \frac{e^{i\omega r}}{r} \tilde{T}(\omega, \hat{\mathbf{n}})$$

where we define the spacetime Fourier transform

$$\tilde{T}(\omega, \hat{\mathbf{n}}) = \int d^3x' e^{-i\omega \hat{\mathbf{n}} \cdot \mathbf{x}'} \tilde{T}^{\mu\nu}(\omega, \mathbf{x}') = \int d^4x' e^{ik_\rho x'^\rho} T^{\mu\nu}(t, \mathbf{x}')$$

and that $k_\rho = (\omega, -\omega \hat{\mathbf{n}})$. Now, we have the Fourier transformed asymptotic distance-independent waveforms in the local wave zone for gravitational waves.

$$\tilde{\kappa}^{\mu\nu}(\omega, \hat{\mathbf{n}}) = r \tilde{\bar{h}}^{\mu\nu} \approx \kappa_g e^{i\omega r} \tilde{T}(\omega, \hat{\mathbf{n}})$$

4 Gravitational Radiation from Strings

For Nambu-Goto strings, the source terms are obtained by varying S with $g_{\mu\nu}$.

$$\begin{aligned}
\delta S &= -\frac{\mu}{2} \int d^2\zeta \eta^{ab} \partial_a X^\mu \partial_b X^\nu \delta g_{\mu\nu}(X) \\
&= -\frac{\mu}{2} \int d^2\zeta \eta^{ab} \partial_a X^\mu \partial_b X^\nu \int d^4x \delta^{(4)}(x - X) \delta g_{\mu\nu}(x) \\
&= \int d^4x \left[-\frac{\mu}{2} \int d^2\zeta \eta^{ab} \partial_a X^\mu \partial_b X^\nu \delta^{(4)}(x - X) \right] \delta g_{\mu\nu}(x) \\
&= -\frac{1}{2} \int d^4x \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu}(x)
\end{aligned}$$

Matching the integrands, the stress-energy tensor in the flat space $\sqrt{-g} = 1$ is

$$\begin{aligned}
T^{\mu\nu}(x) &= \mu \int d^2\zeta \eta^{ab} \partial_a X^\mu \partial_b X^\nu \delta^{(4)}(x - X) \\
&= \mu \int d^2\zeta \left(\dot{X}^\mu \dot{X}^\nu - X'^\mu X'^\nu \right) \delta^{(4)}(x - X)
\end{aligned}$$

Taking the parametrization

$$X^\mu(\tau, \sigma) = \frac{1}{2} \left[X_+^\mu(\sigma_+) + X_-^\mu(\sigma_-) \right], \quad d^2\zeta = d\tau d\sigma = \frac{1}{2} d\sigma_+ d\sigma_-$$

the stress-energy tensor can be expressed using $X_\pm^{(1)}$.

$$T^{\mu\nu}(x) = \mu \int d\sigma_+ d\sigma_- \left(X_+^{(1)\mu} X_-^{(1)\nu} + X_+^{(1)\nu} X_-^{(1)\mu} \right) \delta^{(4)}(x - X)$$

In Fourier space, we can express

$$\begin{aligned}
\tilde{T}^{\mu\nu}(k) &= \int d^4x e^{-ik_\rho x^\rho} T^{\mu\nu} \\
&= \mu \int d\sigma_+ d\sigma_- e^{-ik_\rho X^\rho} \left(X_+^{(1)\mu} X_-^{(1)\nu} + X_+^{(1)\nu} X_-^{(1)\mu} \right) \\
&= \mu \int d\sigma_+ d\sigma_- e^{-\frac{i}{2} k_\rho X_+^\rho} e^{-\frac{i}{2} k_\rho X_-^\rho} \left(X_+^{(1)\mu} X_-^{(1)\nu} + X_+^{(1)\nu} X_-^{(1)\mu} \right) \\
&= \mu \left(I_+^\mu I_-^\nu + I_+^\nu I_-^\mu \right)
\end{aligned}$$

where we define

$$I_\pm^\mu = \int d\sigma_\pm e^{-\frac{i}{2} k_\rho X_\pm^\rho} X_\pm^{(1)\mu} = \int d\sigma_\pm e^{-\frac{i}{2} \omega(\sigma_\pm - \hat{n} \cdot \mathbf{X}_\pm)} X_\pm^{(1)\mu}$$

if we choose $X_\pm^\mu = (\sigma_\pm, \mathbf{X}_\pm)$

Now, we substitute the expansion at X_c^μ , choose $\hat{\mathbf{n}} = \mathbf{X}_{c\pm}^{(1)}$, and gauge out the $X_{c\pm}^{(1)\mu}$ term.

$$\begin{aligned} I_\pm^\mu &\approx \int_{-\infty}^{\infty} d\sigma_\pm e^{-\frac{i}{2}\omega \left[\sigma_\pm - \hat{\mathbf{n}} \cdot \left(\sigma_\pm \mathbf{X}_{c\pm}^{(1)} + \frac{1}{2} \sigma_\pm^2 \mathbf{X}_{c\pm}^{(2)} + \frac{1}{6} \sigma_\pm^3 \mathbf{X}_{c\pm}^{(3)} \right) \right]} \left(X_{c\pm}^{(1)\mu} + \sigma_\pm X_{c\pm}^{(2)\mu} \right) \\ &= \int_{-\infty}^{\infty} d\sigma_\pm e^{-\frac{i}{12}\omega \sigma_\pm^3 |\mathbf{X}_{c\pm}^{(2)}|^2} \sigma_\pm X_{c\pm}^{(2)\mu} \end{aligned}$$

We can obtain an analytic expression for the asymptotic behaviour of this integral when $\omega \rightarrow \infty$ using the saddle point approximation; if $\sigma_\pm - \hat{\mathbf{n}} \cdot \mathbf{X}_\pm \neq 0, \forall \sigma_\pm$, the asymptotic behaviour is exponential decay and if there is a point on the string worldsheet where $\sigma_{pm} - \hat{\mathbf{n}} \cdot \mathbf{X}_\pm = 0$, one obtains a spectrum.

Now, using

$$\int_0^\infty x e^{-ax^3} dx = \frac{1}{3a^{\frac{2}{3}}} \int_0^\infty u^{-\frac{1}{3}} e^{-u} du = \frac{1}{3a^{\frac{2}{3}}} \Gamma\left(\frac{2}{3}\right)$$

we obtain

$$\begin{aligned} I_\pm^\mu &= 2i \operatorname{Im} \left[\int_0^\infty d\sigma_\pm e^{-\frac{i}{12}\omega \sigma_\pm^3 |\mathbf{X}_{c\pm}^{(2)}|^2} \sigma_\pm X_{c\pm}^{(2)\mu} \right] \\ &= \frac{2}{3} i X_{c\pm}^{(2)\mu} \Gamma\left(\frac{2}{3}\right) \operatorname{Im} \left(\frac{i\omega |\mathbf{X}_{c\pm}^{(2)}|^2}{12} \right)^{-\frac{2}{3}} \\ &= -\frac{i\Gamma\left(\frac{2}{3}\right)}{\sqrt{3}} \frac{X_{c\pm}^{(2)\mu}}{|\mathbf{X}_{c\pm}^{(2)}|^{\frac{4}{3}}} \left(\frac{12}{\omega} \right)^{\frac{2}{3}} \end{aligned}$$

Writing the waveform explicitly and neglecting the phase term, we obtain an estimate for the amplitude of the asymptotic waveform.

$$\tilde{\kappa}^{\mu\nu}(\omega) \approx \frac{4G\mu}{\omega^{\frac{4}{3}}} \frac{[12\Gamma\left(\frac{2}{3}\right)]^{\frac{4}{3}}}{3} \frac{X_{c+}^{(2)\mu} X_{c-}^{(2)\nu} + X_{c+}^{(2)\nu} X_{c-}^{(2)\mu}}{|\mathbf{X}_{c+}^{(2)}|^{\frac{4}{3}} |\mathbf{X}_{c-}^{(2)}|^{\frac{4}{3}}}$$

The result yields the asymptotic behavior of the spectrum $\tilde{\kappa}^{\mu\nu} \propto \omega^{-\frac{4}{3}}$ where the direction of the radiation beam coincides with $\mathbf{X}_{c\pm}^{(1)}$.